

Crush Injury

Criteria

- Crush muscle mass
- Prolonged injury of 4-6 hrs (can occur in less time depending on circumstance)
- Compromised local circulation

Pathophysiology of Crush Injury

- Hypovolemic shock 2^o to third spacing
- Metabolic acidosis – Hyperkalemia, hyperphosphatemia, hypocalcemia, rhabdomyolysis, renal failure

Goals of care

- Prevention of acute kidney injury associated with rhabdomyolysis
- Early/generous volume resuscitation may be required to overcome the third spacing created by the injured muscle that may be responsible for hypovolemic shock

Note

- Sodium bicarbonate is no longer uniformly recommended
 - Patients who receive more aggressive fluid management early in their injury are less likely to require dialysis.
 - Patients who receive sodium bicarbonate with less fluid may still require dialysis but also may experience unrecognized electrolyte shifts.
- Some institutions teach placing a tourniquet prior to releasing the pressure – we do not teach this practice unless life threatening bleeding is expected or present.
 - Contact a medical director prior to placing a tourniquet solely for crush injury purposes
 - Do not withhold a tourniquet when life-threatening or serious bleeding is present

EMT/AEMT

- Assess airway
- Assess injury for pulse, motor function, sensation – repeat assessment every 15 minutes
- If possible, notify medical direction of the ongoing rescue operation – this is for awareness in case further needs arise

Paramedic

- If possible, place cardiac monitor to observe for electrolyte associated changes of cardiac rhythm
 - New 1st degree heart block is an early sign of hyperkalemia
 - Widened QRS
 - Biphasic T waves
- Obtain IV or IO proximal to crush site, prefer uninvolved extremity
- Monitor urine output if crush extends past 1 hour
- **Normal saline (preferred) or Lactated Ringers** 1L/hr to achieve 200-300ml/hr urine output
- At two hours, if the patient is producing urine at this rate and is still under the object that is causing the injury, normal saline may be reduced to 0.5L/hr

****If possible, contact medical direction in the event of changes in mental status, EKG changes, or less than 200-300ml/hr urine output****

<https://www.uptodate.com/contents/crush-related-acute-kidney-injury/print>

Smith, Jason MD; Greaves, Ian MD. Crush Injury and Crush Syndrome: A Review. The Journal of Trauma: Injury, Infection, and Critical Care: May 2003 - Volume 54 - Issue 5 - p S226-S230 doi: 10.1097/01.TA.0000047203.00084.94

Sukru Sever M, Vanholder R. Management of Crush Victims in Disasters: Highlights in Recently Published Recommendations. CJASN: Feb 2013, 8(2)328-335; doi: 10.2215/CJN.07340712